

The following is an excerpt from a school assignment. In which, I was to discuss the negative consequences that come with inadequate software testing using a real example.

Health - Patients at the Instituto Oncológico Nacional (ION) in Panama Underwent Complications Related to Radiation Exposure

From August 2000 until March 2001, just under thirty patients experienced excessive gamma ray treatment in Panama's Instituto Oncológico Nacional (ION) [1]. By May of the following year, seventeen patients died from complications related to radiation exposure [2]. The overexposure is attributable to a calculation issue made by the doctors' computerized treatment planning system (TPS) [3]. Doctors did not realize the issue until a significant number of patients started to show side effects to the radiation exposure [1].

The TPS is designed to determine the treatment duration and dosage distribution for each patient [3]. These treatment times are calculated based on the number of shielding blocks doctors enter into the TPS [3]. Doctors are expected to enter each shielding block into the TPS separately and are allowed to enter a maximum of four shielding blocks per field [3].

According to the International Atomic Energy Agency, a radiation oncologist requested doctors to enter five blocks into a field [3]. In effort to comply with the oncologist's request, doctors entered multiple shielding blocks into one block, and in doing so, the TPS instructed the doctors to treat patients for a substantially-longer time than humanly-bearable [3].

The software was not created for doctors to enter more than four shields; however, doctors found a way to bypass the constraint, causing them to treat patients using faulty calculations. This type of bug would be considered a fault, describing a static defect in the software that can lead to failure. Failure describes bugs that negatively affect the software's external behavior. The static defect is the user's ability to bypass the default input entry, and the failure would be the calculation results neither the doctors, software developers, nor software testers accounted for.

The technologists were reported to be arrested for manslaughter as a result of the patients' deaths [4]. The operators at the ION were also penalized by the national healthy ministry [5]. Many complained about the latter charge, claiming the penalty targeted individuals as opposed to the procedures leading to their decisions [5].

References

[1] J. McCormick. "Can Software Kill?" eWeek. <https://www.eweek.com/it-management/can-software-kill/> (August 23, 2024)

[2] Robert Johnston. "Panama City radiotherapy accident, 2000-2001." Database of radiological incidents and related events—Johnston's Archive. <https://www.johnstonsarchive.net/nuclear/radevents/2000PAN1.html> (August 23, 2024)

[3] M. Akashi. et al. "Investigation of an Accidental Exposure of Radiotherapy Patients in Panama." International Atomic Energy Agency. https://www-pub.iaea.org/mtcd/publications/pdf/pub1114_scr.pdf (August 23, 2024)

[4] "Panama charges technologists with manslaughter" Aunt Minnie.com. <https://www.auntminnie.com/clinical-news/radiation-oncology-therapy/article/15558590/panama-charges-technologists-with-manslaughter> (August 28, 2024)

[5] "Panama to review employee sanctions following nuclear accident" Aunt Minnie.com. <https://www.auntminnie.com/clinical-news/radiation-oncology-therapy/article/15558346/panama-to-review-employee-sanctions-following-nuclear-accident> (August 28, 2024)